

Using Scales

One method of measuring weight is to use scales. The faces of these scales are shown in 100 g intervals with larger divisions every 500 g. The kilogram readings have a number assigned. Starting at zero through to 5 kg, the scale travels past 5 kg but doesn't reach 6 kg, the maximum reading being 5.5 kg.

Column 1 asks read the weight displayed on the scales, the readings are pointing at an exact division. So answer in kg, but to one decimal place, ie: readings such as 3.2 kg or 4.1 kg.

Column 2 is the reverse, position the pointers so that they represent the given weights. Note that you are required to approximate the position of the needle as it may have to be positioned in between divisions, (Q.7 and Q.10). It also asks questions in grams.

Column 3 is similar to column 2 except that an addition is required first. The table describes a certain number of 100 g, 250 g and 1 kg weights that are on a particular scale. Add them across the row of the table and then write the total next to the question number, then point that pointer.

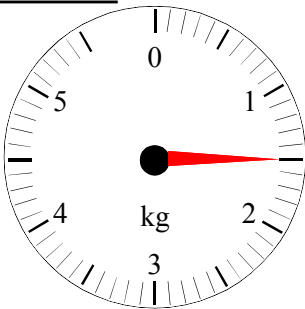
Using Scales

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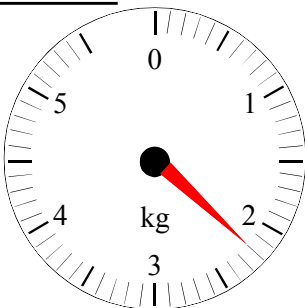


Write the weight shown on the scales below.

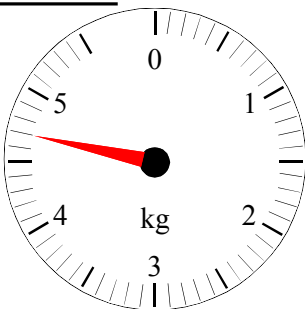
1 kg



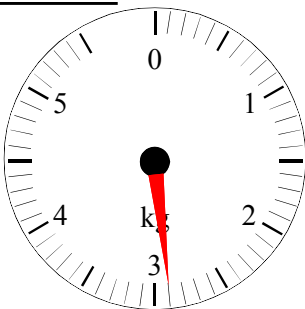
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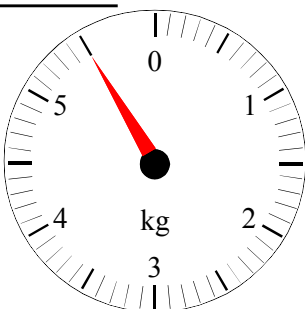
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4

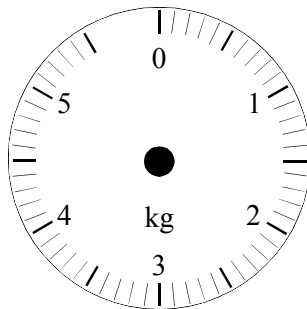


5

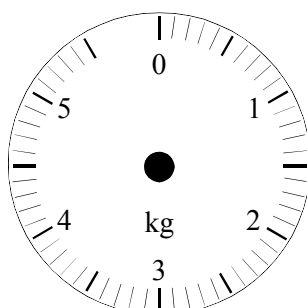


For the weights given draw the pointer in the correct position

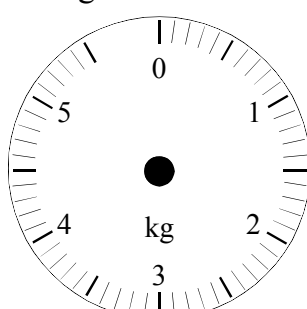
6 4.4 kg



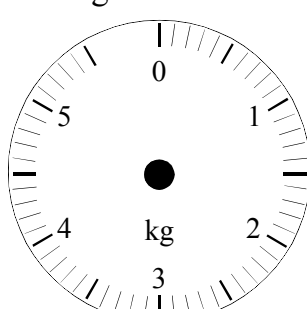
7 2.55 kg



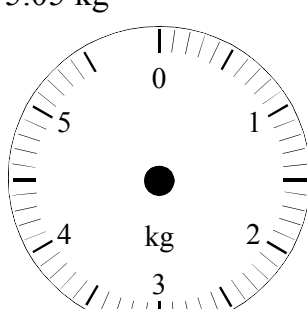
8 900 g



9 3 200 g



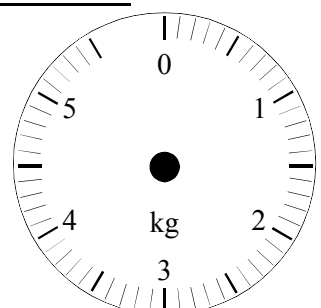
10 5.05 kg



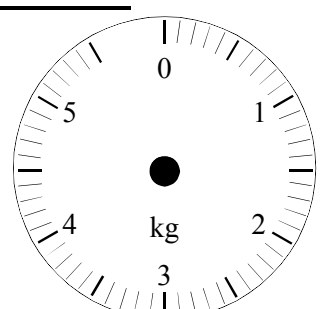
Add the weights in the table below, put the total in the space then point the pointer to the total weight

Scale	100 g	250 g	1 kg
11	3	-	2
12	1	2	3
13	2	-	2
14	1	1	4

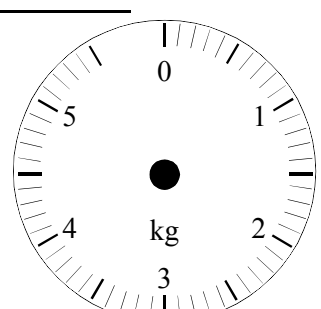
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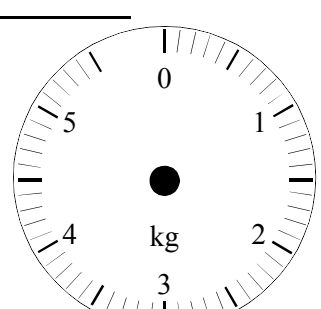
12



13



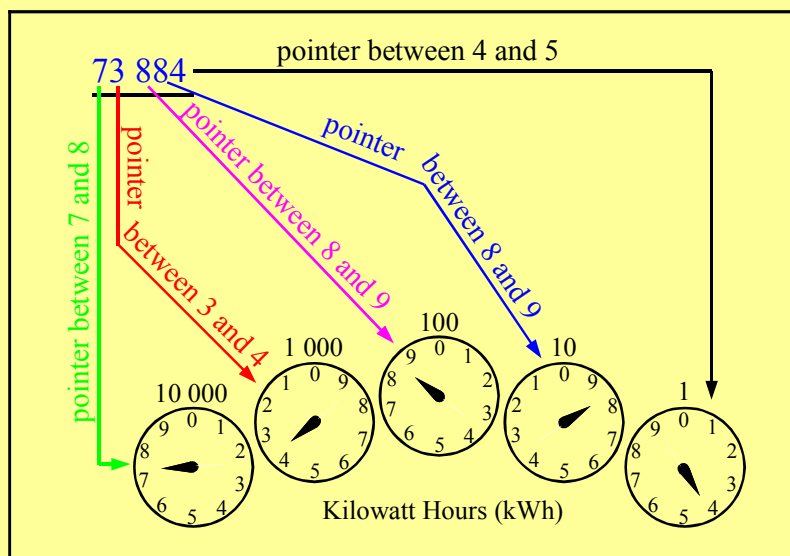
14



Reading Water and Electricity Meters

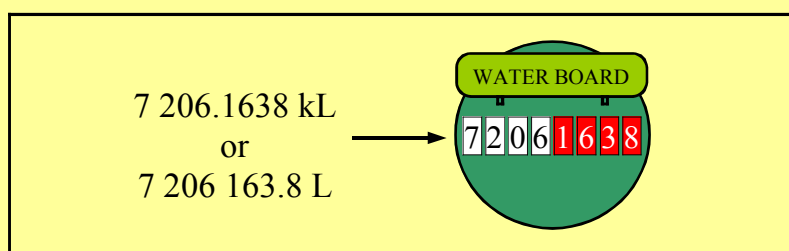
The water and electricity supplied to your home passes through meters which record the amount used. Electricity meters are gear-driven, the scales change between clockwise and anti-clockwise, with the adjacent pointers spinning in opposite directions. There is an easy way to read the scales, just look where the needle is and the number it represents is the smallest of the two numbers the needle is between. Note that zero is both 10 and zero. If the pointer is between the 9 and 0 the smallest number is 9, (zero = 10). If the pointer is between 0 and 1 the smallest number is 0, as zero = 0.

Column 1 starts with 4 questions reading the meters shown. The meters give a 5 digit reading from 10 thousands down to units (1's), the units are in kWh (kilowatt-hours). Read each from left to right and build the five digit number. The next 4 questions ask you to put the pointers in the correct position. Start with the 10 000's first and work your way left to the right.



Column 2 are water meters these are more standard in layout. The meters measure water usage in kL (kilolitres). The end 4 digits are inverted (white on red). These end digits represent 4 decimal places in kL. **If measuring in L, the last digit is tenths of a L (1 d.p).** The first 3 questions compare a 'before' and 'after' reading and then require subtraction to calculate the amount of water used. Note that these meters deal with large units, some numbers are not required in the calculation due to them not changing, a short cut - but be careful!

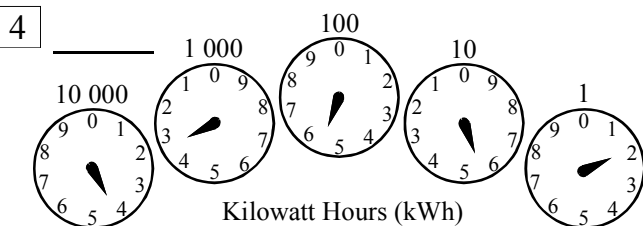
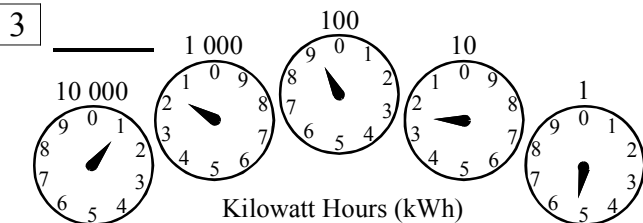
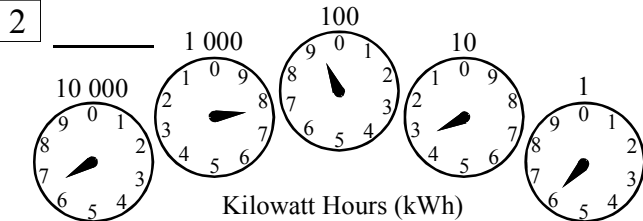
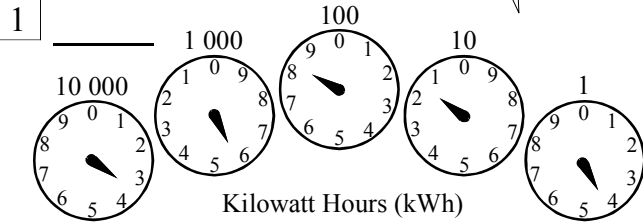
The last 3 questions give the 'before reading' and then require this reading to be added to the usage to get the 'current reading'. After completing the sum, write the answer in the spaces on the meter.



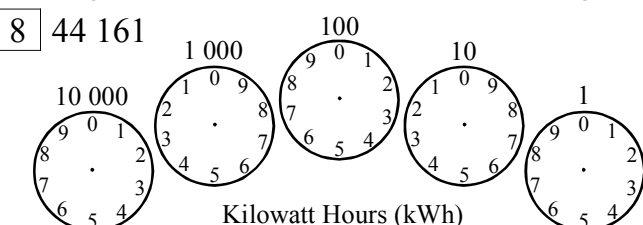
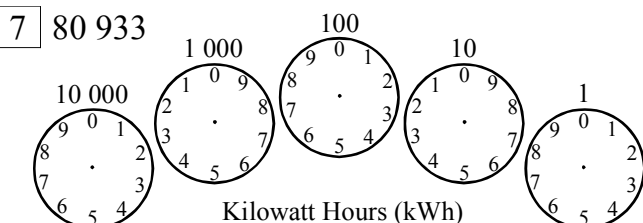
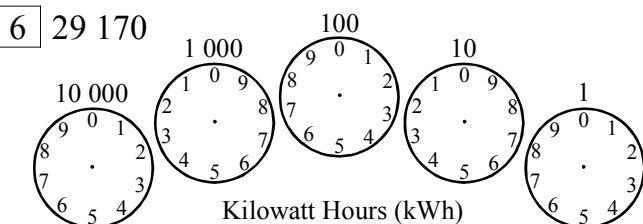
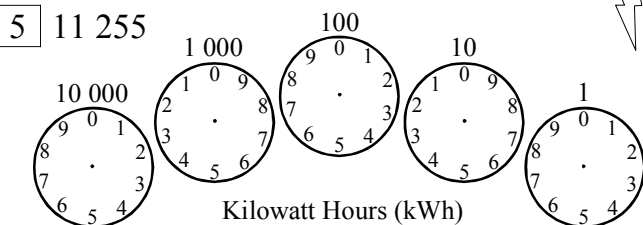
Reading Water and Electricity Meters

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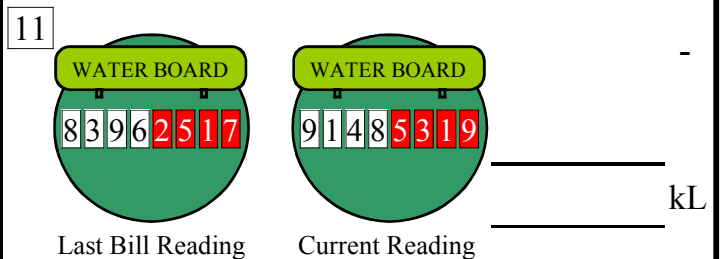
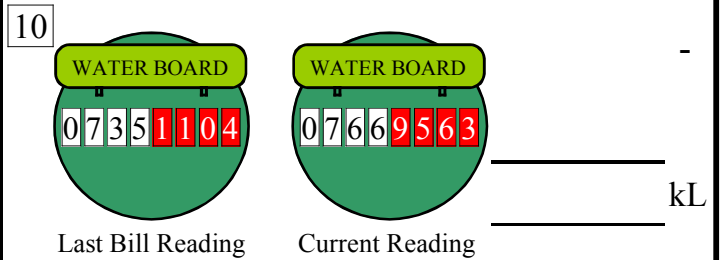
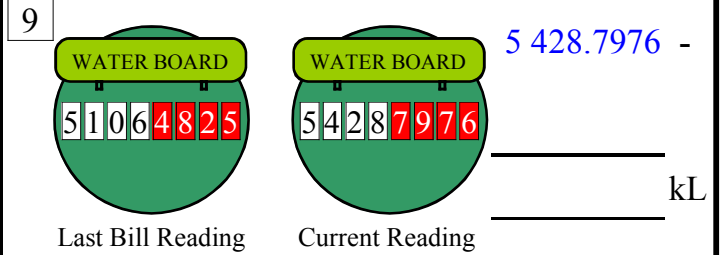
Read the dials below and write the number they represent



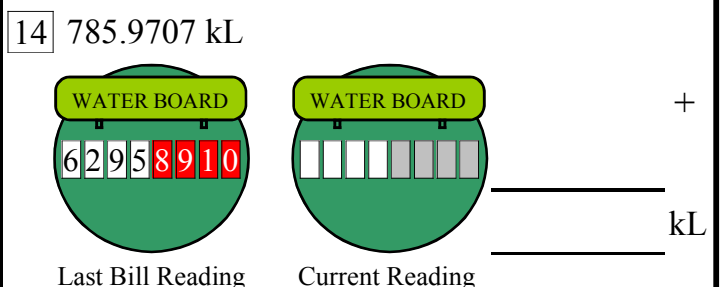
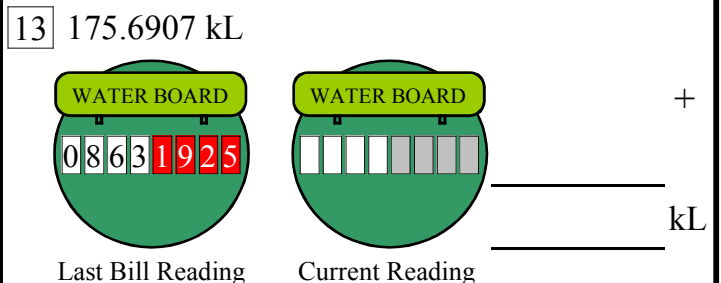
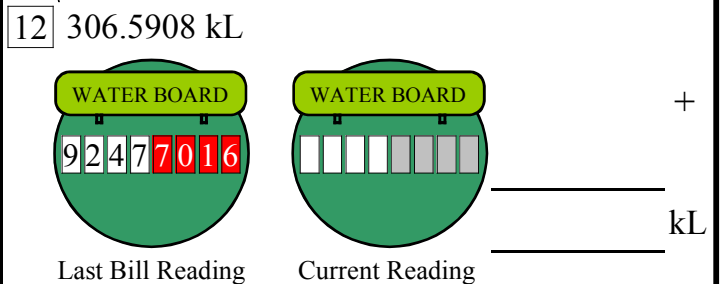
Now the reverse, put in the pointers!



Read the 'last bill' reading then the 'current reading' and calculate the water usage since the last reading.



Given the water used and the last reading add them and write in the current reading that should be on the meter.



Fuel Gauges

Fuel gauges are an indication of how much fuel is in a car's petrol tank. Most commonly fuel gauges have divisions at empty, $\frac{1}{4}$ of a tank, $\frac{1}{2}$ a tank, $\frac{3}{4}$ of a tank and full. In the city, gauges aren't that critical, with 24 hr petrol stations in great numbers. In the country away from the conveniences of the city, the understanding of a fuel gauge is more important.

In column 1 the fuel gauges are shown, put the fraction that the gauge's pointer indicates into the spaces provided. Note that the pointers are meant to be pointing exactly at a division or exactly half way between a division. When the pointer is halfway between divisions the reading will be in 'eighths'. Look at the gauge below, it has the eighths marked.

Column 2 is the reverse, given the fraction draw the pointers to the correct position.

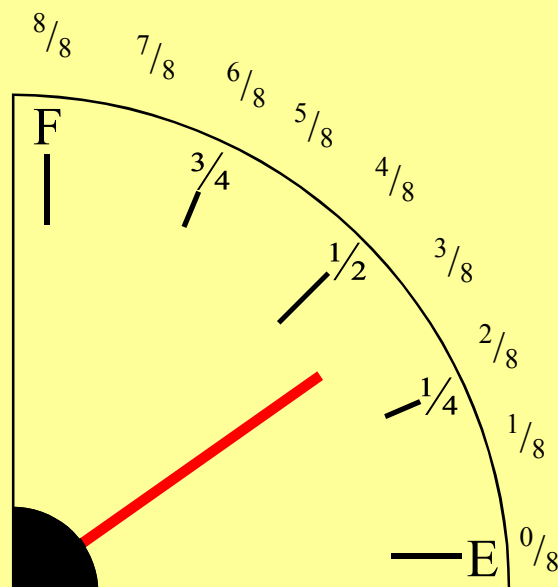
Column 3 asks you to use a fuel gauge to estimate the fuel that remains in the tank as well as the distance that the car can continue until it will run out of fuel. This requires some multiplication of fractions skills, the method is as follows:

- To calculate the litres that remain divide 32 L (tank capacity) by the denominator and multiply by the numerator. This means divide 32 by the bottom number in the fraction, then multiply this number by the top number.
- The same method is used for the distance that the car can still travel. Divide 400 (max km) by the denominator (bottom number in fraction) and then multiply the answer by the numerator (top number in the fraction).

For the gauge below the reading is $\frac{3}{8}$. This means that if a car has a 32 L tank and a maximum travel distance on the tank of 400 km (city), then:

Fuel left is $32 \div 8 \times 3 = 4 \times 3 = 12$ L

Distance until empty is $400 \div 8 \times 3 = 50 \times 3 = 150$ km



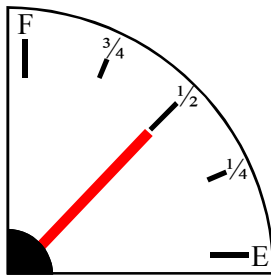
Fuel Gauges

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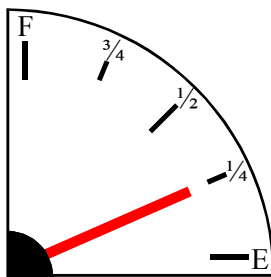
Estimate the fraction of a fuel tank that remains on the gauges below



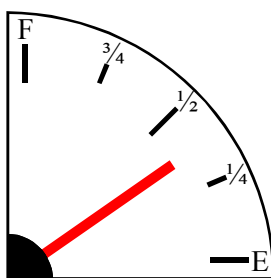
1



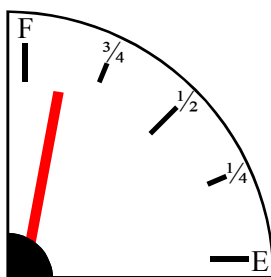
2



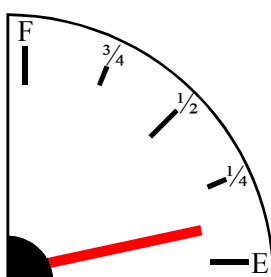
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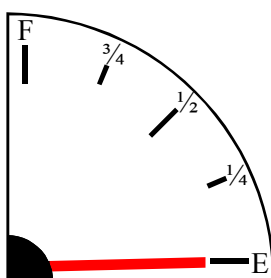
4



5



6

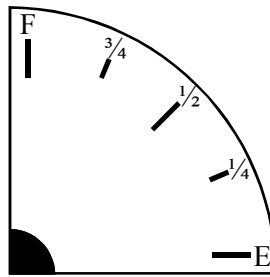


This time the fraction is given, place the needle in the correct position.



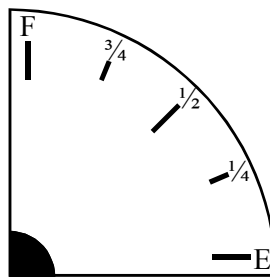
7

$$\frac{3}{4}$$



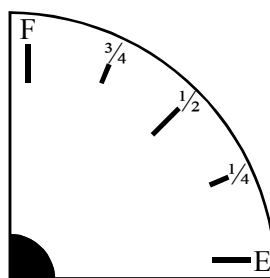
8

$$\frac{7}{8}$$



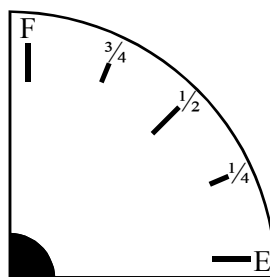
9

$$\frac{1}{4}$$



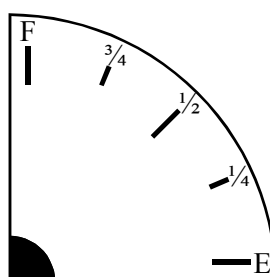
10

$$\frac{3}{8}$$



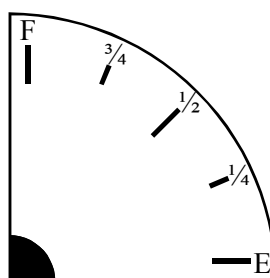
11

$$\frac{1}{2}$$



12

$$\frac{5}{8}$$

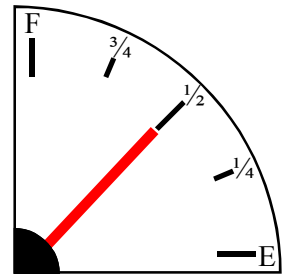


A full tank holds 32 L and travels 400 km (city). Use the gauges to estimate the unused fuel and km that can still be travelled.



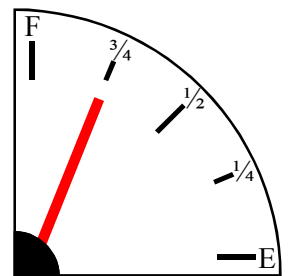
13

L: _____ km: _____



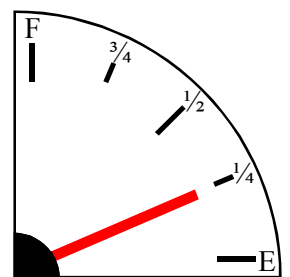
14

L: _____ km: _____



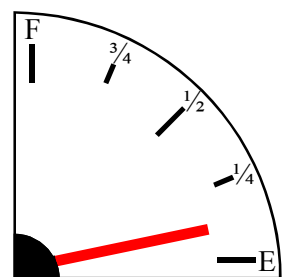
15

L: _____ km: _____



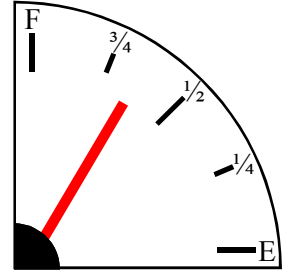
16

L: _____ km: _____



17

L: _____ km: _____



Using a Ruler - Line Measurement

When a line is measured using a ruler the answer can be expressed in mm or cm. If the ruler has both units on it you can use the side needed for the answer. However you should also be able to convert between the two units. You should know that:

$1 \text{ cm} = 10 \text{ mm}$, so $2 \text{ cm} = 20 \text{ mm}$, and decimals $3.8 \text{ cm} = 38 \text{ mm}$ etc.

Column 1 has horizontal lines and vertical lines. It is the thick blue lines you are measuring, don't confuse them with the thin black lines for writing your answer on. The first column asks for answers in cm (to 1 d. p.), that means 3.7 cm, 11.2 cm and so on. The screw and nut at the bottom of the page can be measured either across the object or use the lines with the double-ended arrow between them. The second column is much the same except the lines are angled, note that **measurement for this column is in mm**.

Use the same method with column 3 only write the size of each side and then add them for the total.

Using a Ruler - Line Measurement

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Use a ruler to measure these, express the answer in cm (to 1 d. p.)

1 _____

2 _____

3 _____

4 _____

5 _____

6 _____

7 _____

13 _____

12 _____

11 _____

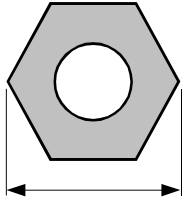
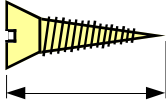
10 _____

9 _____

8 _____

14 _____

15 _____



Measure these angled lines to the nearest mm, express your answer in mm



16 _____

17 _____

18 _____

19 _____

20 _____

21 _____

22 _____



Measure the perimeter of these shapes to the nearest mm, write the side length on each side, then total answer in mm.

23 _____

24 _____

25 _____

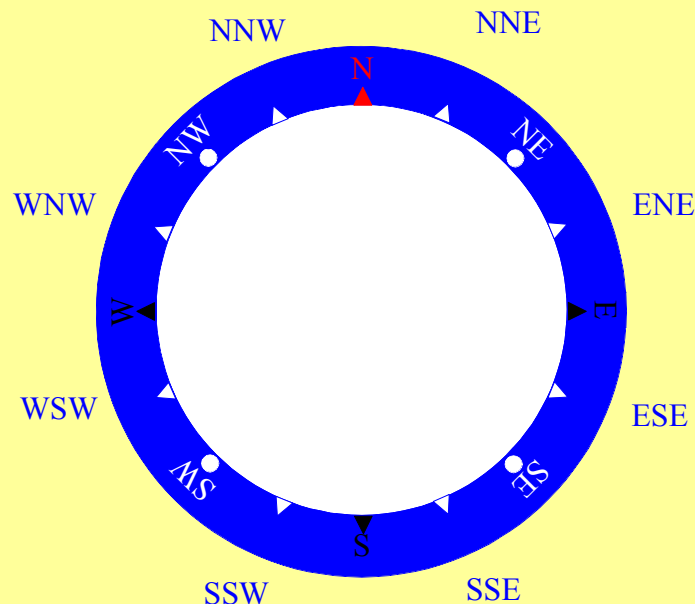
26 _____

Using a Compass

A compass is used as a measure of direction. A compass needle always points to Magnetic North, so on this sheet the red section of the needle is pointing North and the white and red striped end is pointing South. The view is as if you are holding the compass against your chest looking down at it.

The compasses all have the blue ring with N (North) straight up. To find the direction you are facing count how many divisions (triangles and circles) that the red pointer is from North. Then count back the same number of divisions the opposite way from North and that is the direction you are facing.

The readings on this sheet will be one of the readings on the dial below. If you are not familiar with the 16 directions of the compass take the time to look at the dial, as the dials on the sheet don't have all the directions (shown in blue letters) that you need to know.



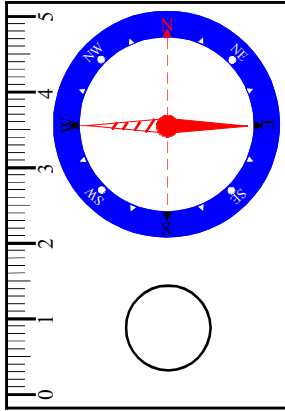
Using a Compass

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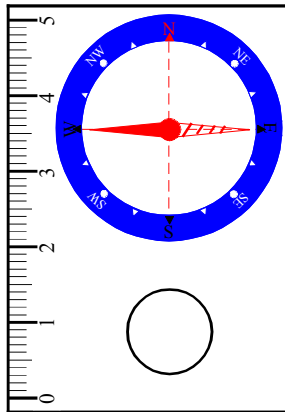


If the compass was held in front of you with the needles as shown which direction are you facing?

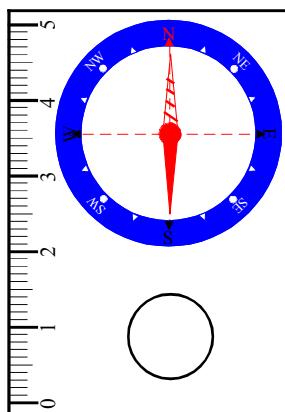
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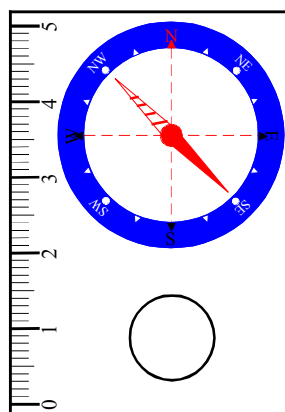
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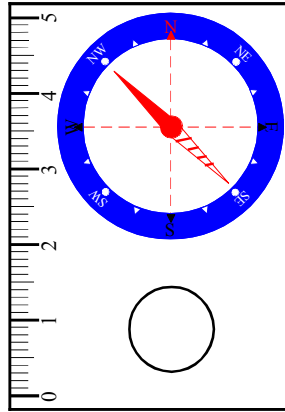
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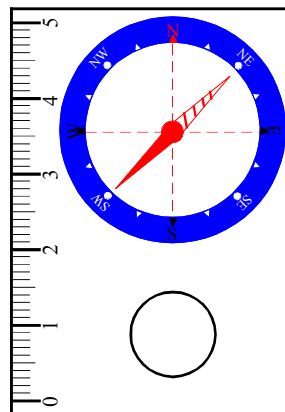
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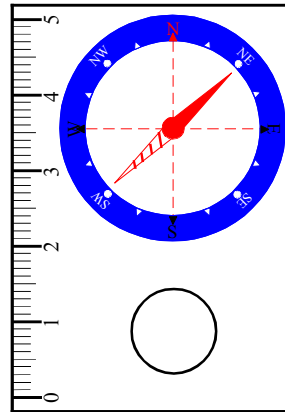
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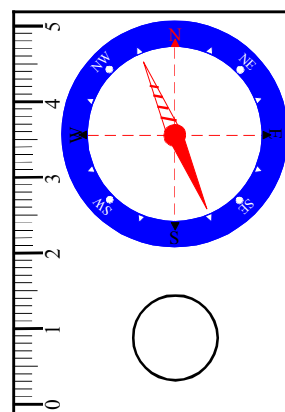
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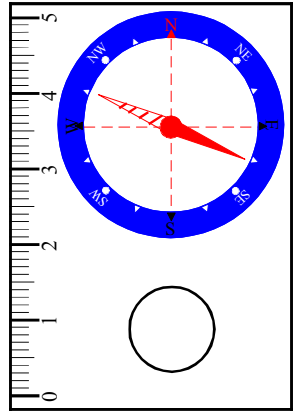
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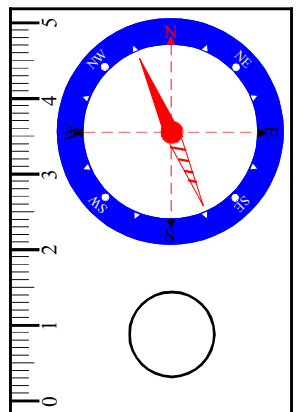
8



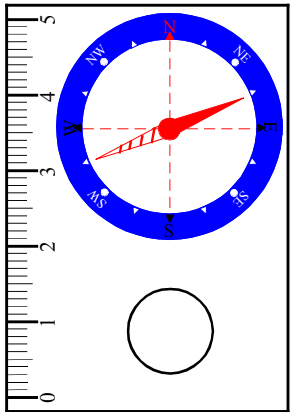
9



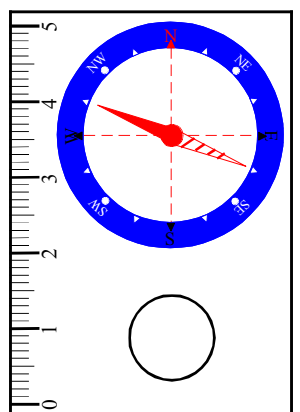
10



11



12



Car Gauges

This sheet asks you to put information on the gauges, a representation of a car's dashboard.

This is the method:

- The fuel gauge is on the left hand side of the dashboard. Draw a needle to the amount of fuel specified. Remember that between $\frac{1}{4}$'s are $\frac{1}{8}$'s of a tank, see the fuel gauge sheet if you need to.
- The speedometer is numbered 0 - 200 km/h. The speed increases by 10 km/h divisions with numbers at every 20 km/h. The odometer is the top set of boxes in the speedometer, while the tripmeter is the bottom set.
- The tachometer is the next large faced gauge. It displays engine speed with divisions at every 500 rpm with numbers at every 1 000 rpm. Note that car manufacturers use '× 1 000' so that single numbers are used. So 1 on the dial is 1 000 rpm. 2.5 would be 2 500 rpm. Note there is an LCD readout to show what gear the car is in.
- The temperature gauge is very simple, it has a green section to show the engine is cold, a blue section to indicate normal temperature and a red section to show if the engine is hot.
- The odometer is the record of the distance traveled during the life of your car it is the 6-digit number in your speedometer. So its largest reading is '999 999' km, then it resets to '000 000'. Write the numbers in the spaces.
- The tripmeter also records the distance traveled but it can be reset. It is often reset when the petrol tank is filled to calculate fuel economy or the distance covered on holidays. It has 4 digits but the last digit is tenths of a km. So its maximum reading is 999.9 km before it resets. Write the numbers in the spaces, note that the last digit is often in inverse (white on black) to show it is tenths of a kilometre.
- Some cars have an LCD readout to show the gear you are in, so that you don't have to look down at the gear shift. Cars usually have 5 gears and some sports cars have 6. Colour the LCD bars.
- When turning at an intersection or overtaking you should use your indicator, colour the arrow, they are usually green or orange.

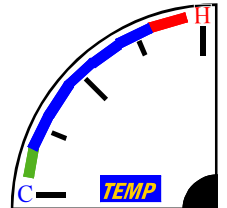
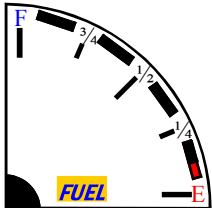
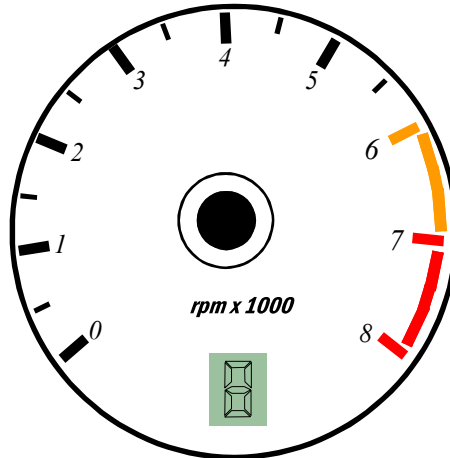
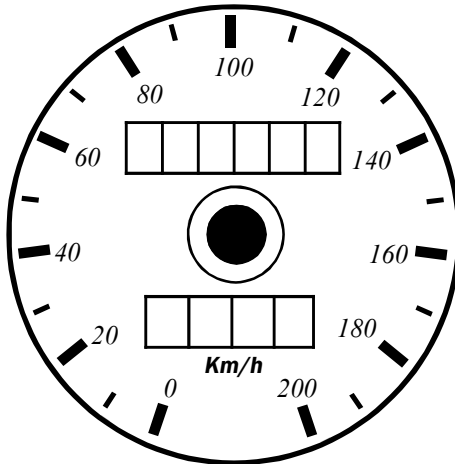
Car Gauges

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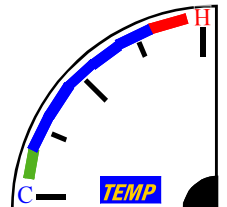
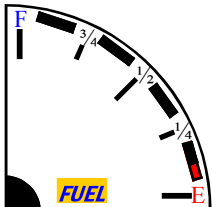
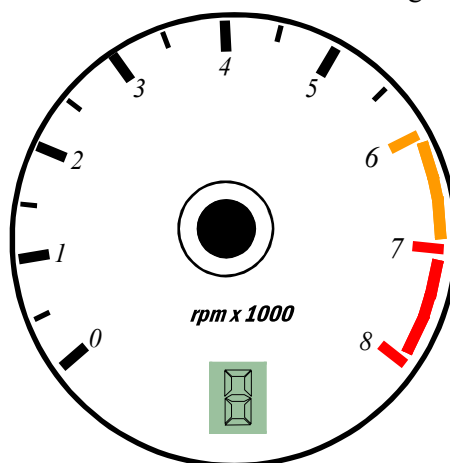
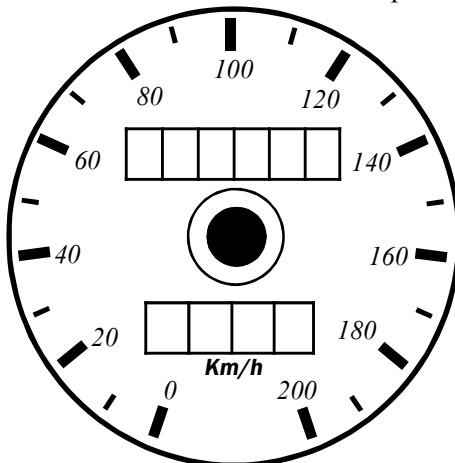


Place all the details on the dashboards, watch out for speed bumps!

1



2



3

